INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

Department of Computer Science and Engineering

MODULE 3 PROJECT REPORT

AI/ML-Based Predictive Maintenance and Vehicle Health Monitoring System

Submitted By

Soumyajit Chatterjee

University Enrollment No.: 12023002001362

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University of Engineering & Management, Kolkata

Kolkata, West Bengal

Executive Summary

This report documents an AI/ML-based predictive maintenance system for connected vehicles. The system analyzes nine vehicle telemetry parameters, predicts failure probability using a tuned Gradient Boosted Decision Tree ensemble, converts the probability into a Vehicle Health Score, and provides ranked diagnostic factors using an explainability layer.

Records	6,000
Normal / Failure	81.23% / 18.77%
Accuracy	96.33%
F1-Score	89.88%
ROC-AUC	0.9854

1. Introduction

1.1 Project Overview

Modern vehicles generate large volumes of telemetry data through sensors connected to CAN bus and OBD-II interfaces. Parameters such as engine temperature, engine speed, oil pressure, vibration, battery voltage, coolant temperature, fuel consumption, vehicle speed, and operating hours provide valuable information about the mechanical and thermal condition of a vehicle.

This project develops an AI/ML-based predictive maintenance and vehicle health monitoring system that analyzes multivariate vehicle telemetry and estimates the probability of vehicle failure. A Gradient Boosted Decision Tree (GBDT) ensemble is used as the primary prognostic model, together with an explainable AI mechanism to identify major telemetry factors contributing to an abnormal condition.

The system produces three principal outputs: failure probability, a Vehicle Health Score (VHS) from 0% to 100%, and ranked contributing diagnostic factors. The architecture is intended for connected-vehicle and fleet-maintenance applications where telemetry can be continuously collected and analyzed.

1.2 Problem Statement

Traditional vehicle maintenance often relies on fixed mileage or operating-hour intervals. Such schedules may fail to capture the actual condition of individual components. Latent thermal, hydraulic, electrical, and mechanical degradation can therefore lead to unexpected failures or unnecessary servicing.

The project addresses the following problem: Given a multivariate vehicle telemetry observation, can a machine learning model accurately identify potential failure states, estimate failure probability, and provide an interpretable indication of the major contributing factors?

1.3 Objectives

- Continuously process multivariate vehicle telemetry.
- Detect normal and potentially failing vehicle states.
- Estimate continuous failure probability, P(fail).
- Convert failure probability into a standardized Vehicle Health Score.
- Classify vehicles into NORMAL, WARNING, and CRITICAL states.
- Identify major telemetry factors contributing to abnormal predictions.
- Provide a foundation for connected-vehicle and fleet-management deployment.

2. Data Preprocessing & Exploratory Data Analysis

2.1 Dataset Description

The project uses a synthetically generated vehicle telemetry corpus containing 6,000 records. The supplied generation log reports a binary class distribution of 81.23% normal records (Class 0) and 18.77% failure/degradation records (Class 1).

The telemetry contains nine principal variables covering thermal, mechanical, electrical, operational, and aging characteristics.

2.2 Telemetry Features

Feature	Unit	Range	Significance
Engine Temperature	°C	60–135	Thermal stress and cooling-system condition
Engine Speed	RPM	600–6500	Engine load and rotational behavior
Oil Pressure	psi	10–65	Lubrication and mechanical health
Vibration	mm/s RMS	0.5–10.0	Mechanical imbalance, bearing wear and misfire
Battery Voltage	V	10.5–14.8	Electrical and charging-system condition
Coolant Temperature	°C	50–125	Cooling-system behavior
Fuel Consumption	L/h	0.5–30.0	Combustion and injector behavior
Vehicle Speed	km/h	0–180	Operating context
Operating Hours	hours	0–10,000	Vehicle age and cumulative usage

2.3 Preprocessing

Missing Value Imputation: Forward filling is used for continuous sensor streams subject to short transmission drops.

Outlier Handling: Telemetry points exceeding approximately ±4σ relative to the operating mode are bounded.

Z-Score Normalization: Features are standardized using $z = (x - \mu) / \sigma$ so that measurements expressed in different units do not dominate downstream processing.

2.4 Exploratory Analysis

The dataset is moderately imbalanced, with normal operation substantially more common than failure/degradation. Consequently, accuracy alone is not sufficient for evaluating the classifier. Failure-class precision, recall, F1-score, ROC-AUC, and the confusion matrix are considered important measures.

The telemetry variables also exhibit physically meaningful interactions. For example, elevated vibration combined with reduced oil pressure and increased engine temperature may indicate a more serious condition than any individual reading alone.

3. Methodology & Modeling

3.1 System Pipeline

Vehicle Sensors → Data Preprocessing → ML Model → Failure Probability → Vehicle Health Score → XAI Attribution → Maintenance Alert

3.2 Model Architecture

The primary prognostic model is a Gradient Boosted Decision Tree (GBDT) ensemble. GBDT builds decision trees sequentially, with later trees reducing errors made by earlier trees. This approach can capture nonlinear relationships and interactions among vehicle telemetry variables without requiring manual interaction engineering.

3.3 Baseline Model

A Random Forest classifier was used as a baseline. The tuned GBDT ensemble achieved higher reported performance across accuracy, precision, recall, F1-score, and ROC-AUC.

3.4 Failure Probability

The classifier produces a posterior probability:

$P(\text{fail}) = P(y = 1 \mid x)$

where $y = 1$ represents the failure/degradation state and x represents the telemetry feature vector.

3.5 Vehicle Health Score

The failure probability is converted into a normalized health score:

$VHS = \max(0, \min(100, (1 - P(\text{fail})) \times 100))$

Thus, higher predicted failure probability corresponds directly to lower vehicle health.

3.6 Maintenance Thresholds

VHS > 70% ($P(\text{fail}) < 0.30$): NORMAL — routine operational profile.

30% ≤ VHS ≤ 70% ($0.30 \leq P(\text{fail}) \leq 0.70$): WARNING — incipient degradation detected; schedule service inspection.

VHS < 30% ($P(\text{fail}) > 0.70$): CRITICAL — high failure risk; immediate intervention recommended.

3.7 Explainable AI

For a sample x, the local attribution of feature j is calculated as:

$A_j = |(x_j - \mu_j) / \sigma_j| \times I_j$

where I_j is the model's global feature importance. Features are sorted by attribution score and the top three factors are reported as the principal diagnostic contributors.

4. Evaluation Metrics & Results

4.1 Validation

The model was evaluated using stratified five-fold validation on the synthetic telemetry corpus. The supplied validation output reports 96.33% accuracy, 89.88% F1-score, and 0.9854 ROC-AUC. The project report additionally provides 91.3% failure-class precision and 88.9% failure-class recall.

4.2 Performance Comparison

Metric	Random Forest	Tuned GBDT
Accuracy	94.8%	96.3%
Precision	88.2%	91.3%
Recall	84.1%	88.9%
F1-Score	86.1%	89.9%
ROC-AUC	0.967	0.985

4.3 Confusion Matrix

The supplied confusion matrix is:

[[956, 19],
 [25, 200]]

This corresponds to 956 true negatives, 19 false positives, 25 false negatives, and 200 true positives. The supplied report identifies the false-positive rate among nominal records as approximately 1.95%.

4.4 Interpretation

The 96.33% accuracy indicates a high proportion of correct classifications. The 91.3% precision indicates that most records flagged as failure are actually failure/degradation cases, while the 88.9% recall indicates that the model detects most of the observed failure cases. The 0.9854 ROC-AUC exceeds the specified target of 0.95.

5. Diagnostic Case Studies

5.1 Scenario 1 — Imminent Thermal and Mechanical Degradation

Input telemetry:
- Engine Temperature: 114.5°C
- Vibration: 5.42 mm/s
- Oil Pressure: 15.2 psi
- Coolant Temperature: 105.8°C

Output:
- Vehicle Health: 4%
- Failure Probability: 96%
- Status: CRITICAL

Major contributing factors:
1. High structural vibration
2. High engine temperature
3. Abnormal oil pressure

The supplied project interpretation associates this combination with severe thermal degradation and reduced oil pressure, potentially producing increased mechanical friction and vibration.

5.2 Scenario 2 — Incipient Degradation

Input telemetry:
• Engine Temperature: 99.2°C
• Vibration: 3.85 mm/s
• Oil Pressure: 26.5 psi

Output:
• Vehicle Health: 72%
• Failure Probability: 28%

Major contributing factors:
1. High structural vibration
2. High engine temperature
3. Abnormal oil pressure

The mathematical output is internally consistent because (1 − 0.28) × 100 = 72%. However, the supplied source narrative describes this case as WARNING/incipient wear, while the formal threshold table classifies VHS above 70% as NORMAL. This report preserves that source-level inconsistency rather than silently changing it.

5.3 Scenario 3 — Nominal Operating Profile

Output:
• Vehicle Health: 99%
• Failure Probability: 1%
• Status: NORMAL

Major contributing factors:
1. Engine speed instability (RPM)
2. Low battery voltage
3. Vehicle speed variance

The result is consistent with the health-score equation: (1 − 0.01) × 100 = 99%.

6. Deployment Strategy

6.1 Proposed Architecture

The intended deployment pipeline is:

CAN/OBD-II Sensors → Edge Telematics Unit → MQTT/Message Broker → Cloud Inference → Time-Series Database → Maintenance Dashboard

The edge unit can ingest high-frequency CAN frames, perform filtering and rolling statistical aggregation, and transmit condensed telemetry over a cellular connection. The cloud inference layer can execute the GBDT model, calculate failure probability and VHS, generate XAI factor rankings, persist historical states, and dispatch maintenance alerts.

6.2 Future Model Maintenance

The supplied architecture proposes periodic retraining as new operational data becomes available. Such retraining can help account for component aging, changing operating conditions, and new vehicle populations.

7. Conclusion

7.1 Summary

This project demonstrates an end-to-end machine learning approach for predictive vehicle maintenance and health monitoring using multivariate telemetry data.

A synthetic telemetry corpus containing 6,000 records was used to develop and validate a binary failure-classification system. The tuned GBDT ensemble achieved 96.33% accuracy, 91.3% failure-class precision, 88.9% failure-class recall, 89.88% F1-score, and 0.9854 ROC-AUC.

The system extends beyond binary classification by converting model output into a 0–100% Vehicle Health Score and adding an explainable AI layer that identifies the principal telemetry factors associated with abnormal predictions.

The architecture provides a foundation for moving from fixed scheduled maintenance toward condition-based and predictive maintenance. Future development could include validation on real-world fleet telemetry, temporal models for sequential sensor behavior, probability calibration, automated maintenance recommendations, and continuous post-deployment model monitoring.